OOP WEEK ASSIGNMENT NO 1 EXERCISES

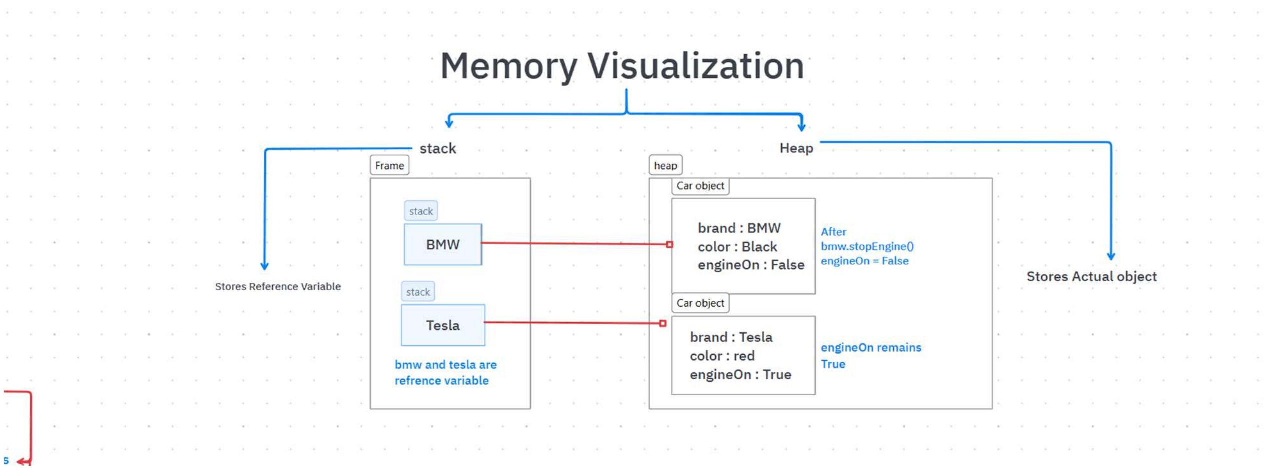
Exercise-01

5 Objects with Attributes and Methods

no	Favourite Object	Relevant Attributes (What it has)	Behaviours (What it can do)
1	Sports Car	brand, model, color, topSpeed, fuelLevel	startEngine(), accelerate(), brake(), drift(), turboMode()
2	Smart Water Bottle	brand, capacity, waterLevel, temperature, color	fillWater(), drinkWater(), remindToDrink(), displayWaterLevel(), cleanBottle()
3	Smart Backpack	brand, color, weight, batteryLevel, GPS	openBag(), lockBag(), chargePhone(), trackLocation(), storeItems()
4	Coffee Machine	brand, waterLevel, coffeeBeans, temperature, powerStatus	makeCoffee(), heatWater(), brewCoffee(), cleanMachine(), turnOff()
5	Smartphone	brand, model, batteryLevel, storage, camera	makeCall(), sendMessage(), takePhoto(), browseInternet(), charge()

Exercise-02

Stack and Heap Diagram For Objects



Exercise-03

Java Varargs

Objective

The purpose of this exercise is to improve the calculator by allowing it to add any number of values using Java Varargs.

Code

```
public double add(double... numbers) {
    double total = 0;

    for (double number : numbers) {
        total += number;
    }

    return total;
}
```

Key Points

- `double... numbers` is called **Varargs**.
- It allows the method to accept any number of values.
- The `numbers` parameter behaves like an array.
- A **for-each loop** is used to calculate the total.
- The method returns the sum of all the given numbers.

Conclusion

This approach makes the calculator more flexible and efficient. Instead of creating different methods for different numbers of inputs, a single method can handle all cases, making the code shorter, cleaner, and easier to maintain.